* One Hot Encoding:

Vocabulary size: 7

Vocabulary {unique words}

Text	O/P
D_1 The food is good	1
D_2 The food is bad	0
D_3 Pizza is Amazing	1

The	food	is	good	bad	Pizza	Amazing
1	0	0	0	0	0	0
0	1	0	0	0	0	0

Test [Burger is bad] → Not able to predict on new data. Burger is not present in the vocabulary

D_1 [[1 0 0 0 0 0 0], [0 1 0 0 0 0 0], [0 0 1 0 0 0 0], [0 0 0 1 0 0 0]] 4×7

D_2 [[1 0 0 0 0 0 0], [0 1 0 0 0 0 0], [0 0 1 0 0 0 0], [0 0 0 0 1 0 0]] 4×7

D_3 [[0 0 0 0 0 1 0], [0 0 1 0 0 0 0], [0 0 0 0 0 0 1]] 3×7

* Advantage and Disadvantage of OHE:

Advantage:

i. Easy to implement with python.
[Sklearn → OneHotEncoder,
Pandas → pd.get_dummies()]

Disadvantage:

i. Sparse Matrix → Overfitting.

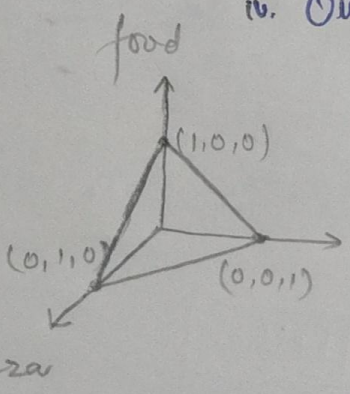
ii. ML-Algorithm → fixed sized i/p
Not present in OHE

iii. No semantic meaning is getting captured.

iv. Out of vocabulary (OOV)

food pizza burger.

[1 0 0]
[0 1 0]
[0 0 1]



→ The distance is almost similar and cannot be distinguished.